

Saksham Dhuria

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EDUCATION

University of Wisconsin Madison

Madison, WI

B.S. in Computer Science(Honors) and Mathematics; GPA - 3.83

May 2027

Relevant Coursework - Operating Systems, Algorithms, Graduate HPC, Data Structures, Stochastic Processes, Real Analysis, Game Theory, Machine Learning, Probability, Numerical Linear Algebra, Object Oriented Programming

EXPERIENCE

OpenAI

San Francisco, CA

Incoming Member of Technical Staff Intern in Fall 2026

Sept 2026 – Dec 2026

Databricks

Sunnyvale, CA

Software Engineer Intern

May 2026 – Present

- Building a scalable data lineage and observability framework in Scala for Apache Spark to improve query tracking, cross-system visibility, and monitoring reliability across distributed data platforms

NVIDIA

Santa Clara, CA

Deep Learning Compiler Engineering Intern

Jan 2026 – May 2026

- Built Drongo, a metamorphic compiler-verification framework that uses Google Centipede to fuzz TileIR compiler by mutating MLIR tests while preserving expected program behavior
- Detected and reported 27 TileIR bugs, including ICEs, functional mismatches, and regression edge cases
- Filed a patent for a compiler-agnostic AI pipeline that converts natural-language mutation ideas into verified mutators, reducing development time from 2 days to 30 mins (~ 97%)

Machine Learning Research Group, UW Madison

Madison, WI

Research Assistant

Jan 2026 – May 2026

- Implementing diffusion models to study hallucinations by disturbing conditioning signals, guidance strength, and sampling trajectories, analyzing how uncertainty and control failures emerge during generation.
- Developing hallucination-aware diffusion pipelines by adding controlled guidance mechanisms, evaluating trade-offs

Capital One

Chicago, IL

Software Engineer Intern

Jun 2025 – Aug 2025

- Increased self-service efficiency for non-technical teams by building FMS, a full-stack intern automation platform
- Enabled real-time updates and secure, schema-less data storage using React, Tanstack, AWS S3, and DynamoDB

Cadence Design Systems

San Jose, CA

Software Engineering Intern

May 2024 – Dec 2024

- Achieved a 7× speedup in dynamic graph computation by implementing a multithreaded C++ scheduler
- Improved code generation accuracy by linking an LLM to TCL via Python APIs and RLHF-based testing

Google Summer of Code: Taskflow

San Jose, CA

Software Engineering Intern

Apr 2024 – Aug 2024

- Boosted cluster throughput by 10% via cache-aware scheduling to improve locality in a distributed C++ system.
- Reduced scheduling latency by improving a work-stealing algorithm with worker-specific alerts for task distribution

PROJECTS

RoboCuber | Arduino, Embedded C, Kociemba Algorithm, Robotics, Backtracking, Pruning Tree

- Coded advanced Kociemba solving algorithms with pruning tables and backtracking for efficient cube solving
- Designed and built a Rubik's Cube robot using Arduino and embedded C for precise physical manipulation

Forecasting Bitcoin Volatility with GARCH Modeling | Python, NumPy, Time Series Analysis, GARCH

- Built a GARCH(1,1) model to forecast Bitcoin price volatility using log returns and conditional variance

TECHNICAL SKILLS

Programming Languages: Python, C++, Java, Golang, C, CUDA, Matlab, JavaScript

Libraries: NumPy, Pandas, TensorFlow, OpenMP, Intel Threading Building Blocks (TBB)

Tools & Frameworks: AWS (S3, DynamoDB), React, GitHub, Perforce, Slurm, Taskflow

Mathematical & Competitive Background: AIME (4× Qualifier), ICPC, Codeforces, AtCoder, USACO